ICSE Class X Chemistry

Chapter-wise Question Bank: Chapters 1–3

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How to use this: Attempt every question cold on paper. Mark only after finishing a section. Log each wrong or blank item as a gap, rewrite it, and retry it after 24 hours. Source tags: **T23–T26** = board paper of that year; **S27 / S26 / S24** = specimen paper of that year. Items without a tag come from the syllabus and are board-style.

Chapter 1: Periodic Properties and Variations of Properties

A. Theory and reasoning

1.1 Define atomic size, metallic character, non-metallic character, ionisation potential, electron affinity and electronegativity.

1.2 State how each of the six properties varies down a group and across a period. Explain each trend using nuclear charge and number of shells.

1.3 Why does atomic size decrease across a period but increase down a group?

1.4 Why does ionisation potential decrease down a group? [T23]

1.5 Why do noble gases have zero electron affinity and the highest ionisation potential in their period? [T24, T23]

1.6 Why are alkali metals strong reducing agents, and why does their reactivity increase down the group?

1.7 Why are halogens strong oxidising agents, and why does their reactivity decrease down the group?

1.8 Why is fluorine the most electronegative element? Why does electronegativity increase across a period?

1.9 Why is the Modern Periodic Law based on atomic number and not atomic mass?

B. Objective questions

1.10 An atom of X forms an ion X^{2+}. The atomic number of X is: (a) 16 (b) 10 (c) 12 (d) 14 [T26]

1.11 Assertion (A): The tendency of losing electrons increases down the group. Reason (R): The most reactive metal is placed at the top of Group 1. Choose: (a) both true, R explains A (b) both true, R does not explain A (c) A true, R false (d) A false, R true [T25]

1.12 Which pair of ions has the same electronic configuration? Cl^-, Li^+, Al^{3+}, K^+ [T25]

1.13 X, Y, Z are successive metals in the same group and X has the smallest atomic radius. Which is the best reducing agent? [T25]

1.14 In the 2nd period, neon has maximum ionisation potential because: (a) unstable electronic configuration (b) it easily accepts electrons (c) it easily loses electrons (d) its outermost shell is completely filled [T24]

1.15 Electron affinity is maximum in: (a) Mg (b) Ar (c) Li (d) Br [T24]

1.16 Element with the largest atomic radius among C, N, Li, Be: [T23]

1.17 An element in period 3 whose electron affinity is zero: (a) neon (b) sulphur (c) sodium (d) argon [T23]

1.18 Which has the largest atomic radius: Al, S, Mg or Na? [S26]

1.19 A → A^{3+} and B → B^{-}. How many electrons are in the outermost shell of atoms A and B? [S24]

1.20 The element with atomic number ___ forms an acidic oxide: 3, 17, 11 or 13. [S24]

1.21 An element in the third period and sixteenth group has how many valence electrons? [S24]

1.22 X and Y are diatomic elements and X is less reactive than Y. Identify X and Y from: (a) X chlorine, Y iodine (b) X fluorine, Y nitrogen (c) X iodine, Y bromine (d) X oxygen, Y nitrogen [S27]

1.23 Statement I: Halogens are very reactive. Statement II: Halogens have high electron affinity. Are both true? [S27]

C. Short answer and application

1.24 P, Q and R are in the same period. P readily loses its one valence electron. Q shares electrons but does not form ions easily. R has high electronegativity. Which is (a) most difficult to reduce, (b) smallest in atomic radius? (c) Arrange P, Q, R in decreasing order of ionisation potential. [T26]

1.25 Name (a) the most electronegative element of period 2, (b) the largest atom of period 3. [T26]

1.26 From Li, K, Ca, F identify the element with (a) the least atomic radius, (b) two valence electrons, (c) the highest electropositive character. [T26]

1.27 Name the element with the highest ionisation potential in each set: (a) Al, Cl, Mg (b) Ne, O, F [T25]

1.28 Atomic numbers of X and Y are 14 and 8. State (a) the period of X, (b) the formula of the compound formed between X and Y. Do not identify the elements. [T25]

1.29 Arrange as instructed: (a) C, F, Be in decreasing atomic size (b) K, Li, Na in increasing ionisation potential [T24]

1.30 Identify: (a) the period-1 element that can be placed in both Group 1 and Group 17, (b) the element with configuration 2, 8, 6, (c) the most electronegative element of period 3. [T24]

1.31 Arrange as instructed: (a) Al, K, Mg, Ca in decreasing reactivity (b) N, Be, O, C in increasing non-metallic character (c) P, Si, F, Be in decreasing number of valence electrons [T23]

1.32 Element P has 3 valence electrons in its L shell and element Q has 7 valence electrons in its M shell. (a) Give the group and period of Q. (b) Is the atomic radius of P more or less than that of Q? [S27]

1.33 Z is a metal above hydrogen in the reactivity series. Its oxide ZO reacts with water to form a hydroxide. State (a) the number of valence electrons in Z, (b) its group, (c) the formula of its hydroxide. [S27]

1.34 The electron affinity of X is greater than that of Y. Compare (a) their oxidising power, (b) their electronegativity, (c) say whether X lies to the left or right of Y in the period. [S26, S24]

1.35 L, M and N have atomic numbers 13, 7 and 10. Which element (a) combines with hydrogen to give a gas that forms dense white fumes with conc. HCl, (b) has zero electron affinity, (c) can form an ionic compound with oxygen? [S26]

1.36 An element has one electron in its outermost shell. Will it have the largest or the smallest atomic size in its period? [S24]

Chapter 2: Chemical Bonding

A. Theory and structures

2.1 Define electrovalent bond, covalent bond, coordinate bond and lone pair. [T25, S24]

2.2 Draw electron dot structures for the formation of NaCl, $MgCl_2$ and CaO.

2.3 Draw electron dot structures of H_2, Cl_2, O_2, N_2, CH_4 and CCl_4. [T24, T23]

2.4 Draw electron dot structures of HCl, NH_3 and H_2O. Explain why they are polar using electronegativity difference.

2.5 Explain the formation of H_3O^+ and NH_4^+ using the lone pair of oxygen and nitrogen. Draw dot diagrams and label the coordinate bond. [T23, S27, S26, S24]

2.6 Draw dot and cross structures of: (a) the ionic compound formed when Mg reacts with dil. HCl, (b) the covalent compound formed when H_2 reacts with Cl_2, (c) the positive ion produced when ammonia dissolves in water. [T25]

2.7 A compound H_2P is commonly used in the laboratory, where H is hydrogen and P a non-metal. Assume P is sulphur and draw the dot and cross structure. [S27]

2.8 Compare electrovalent and covalent compounds in state of existence, melting and boiling points, conductivity, and solubility in water and organic solvents.

2.9 Give reasons: (a) inert gases do not form ions, (b) covalent compounds have low melting and boiling points. [T24]

2.10 Why do ionic compounds not conduct electricity in the solid state but conduct when molten or in solution? [T23]

2.11 Why is CCl_4 non-polar even though each C–Cl bond is polar?

2.12 Why do ionic compounds have high melting points? (Assertion-reason form: A: ionic compounds have high m.p.; R: strong electrostatic attraction between oppositely charged ions.) [S27]

B. Objective questions

2.13 Which pair of elements forms an ionic bond? P: Group 1 and 2, Q: Group 14 and 16, R: Group 2 and 17, S: Group 15 and 18 [T26]

2.14 Highest melting point among methane, sodium chloride, ammonia and ethanol: [T26]

2.15 Polar covalent compound: HCl or CCl_4? [T26] Non-polar covalent molecule: CCl_4 or HCl? [T24]

2.16 Non-polar covalent compounds are (good / bad) conductors of heat and electricity. [T23]

2.17 From CCl_4, PbO, NaCl, CuO, NH_4Cl choose the non-electrolyte. [T23]

2.18 In a water molecule the oxygen atom has: (a) one shared pair (b) three shared pairs (c) two lone pairs (d) one lone pair [T24]

2.19 Which ion contains one lone pair of electrons: OH^-, H_3O^+, NH_4^+, H^+? [S27, S26]

2.20 ___ conducts electricity by the movement of ions: molten iron or molten sodium chloride. [T25]

2.21 Match: Water, Alkali metal, Halogen, Calcium oxide, Weak acid with: lithium, iodine, covalent compound, acetic acid, ionic compound. [T24]

C. Short answer and application

2.22 Atom P has configuration 2, 8, 1 (atomic weight 23) and atom Q has 2, 6 (atomic weight 16). Give the formula of the compound they form and the mass of one mole of it. [S27]

2.23 True or False, with justification: (a) in an electrovalent compound the cation attains the configuration of the noble gas that comes after it in the periodic table, (b) in compound PQ_2 where P gives one electron to each Q atom, PQ_2 is a good conductor of electricity. [S24]

2.24 Why are the hydronium ion and the ammonium ion able to form even though H^+ has no electrons?

2.25 State the characteristic properties of electrovalent compounds, linked to electrolysis (dissociation in molten state and in solution).

Chapter 3: Study of Acids, Bases and Salts

A. Theory

3.1 Define acid, base, alkali and salt in terms of ions produced. Write ionisation equations for HCl, H_2SO_4, CH_3COOH, NaOH, KOH and NH_4OH, and dissociation of NaCl and $CuSO_4$.

3.2 Define basicity. Give one monobasic, one dibasic and one tribasic acid. Differentiate acetic acid and sulphuric acid by the number of replaceable hydrogen ions per molecule. [T26, S24]

3.3 Why can hydrochloric acid not form an acid salt? Why does acetic acid not form an acid salt with a basic radical? [T26, T23, S24]

3.4 Explain the pH scale. Give the pH ranges for acidic, neutral and alkaline solutions and the colours of universal indicator.

3.5 Give the colours of litmus, methyl orange and phenolphthalein in acidic and alkaline solutions. Which indicator turns pink in alkali? Which turns pink in acid? [T23, S24]

3.6 Define normal salt, acid salt, basic salt and complex salt with one example each. Name the salt formed by partial replacement of the hydroxyl group of a di-acidic base by an acid radical. [T24, T25, T23]

3.7 State True or False: A salt is formed by partial or complete replacement of the hydrogen ion of an acid by a metal or electropositive ion. [T26]

3.8 Assertion (A): Dilute H_2SO_4 is a stronger electrolyte than concentrated H_2SO_4. Reason (R): Dilute H_2SO_4 has a higher concentration of mobile ions. [T26]

3.9 Why is concentrated H_2SO_4 nearly non-conducting while dilute H_2SO_4 conducts well?

B. Action of dilute acids on salts and metals

3.10 Write balanced equations, name the gas evolved and give its test, for dilute HCl or H_2SO_4 acting on: (a) sodium carbonate, (b) sodium hydrogen carbonate, (c) sodium sulphite, (d) sodium hydrogen sulphite, (e) ferrous sulphide or zinc sulphide.

3.11 Compound X with dilute HCl releases a gas that turns moist lead acetate paper black and decolourises acidified $KMnO_4$. Identify X. Options: sodium nitrate, sodium sulphide, sodium carbonate, sodium sulphite. [S27]

3.12 Compound X with dilute H_2SO_4 releases a gas that turns acidified $K_2Cr_2O_7$ orange to green. Is X lead nitrate, copper carbonate, sodium chloride or potassium sulphite? [S26]

3.13 Which gas decolourises potassium permanganate: SO_2, NH_3, HCl or CO_2? [T25]

3.14 Rita added dil. HCl to copper, iron, magnesium and zinc and noted: copper gas, iron gas, magnesium no gas, zinc gas. Which observations are correct? [T26]

3.15 Give one relevant observation when dil. HCl is added to iron(II) sulphide. [T26]

C. pH and indicator application

3.16 A student tests P, Q, R, S with pH paper and gets blue, green, red, violet. Which is neutral? [S27]

3.17 Solutions X, Y, Z have pH 2, 7, 13. Which will (a) liberate SO_2 when heated with sodium sulphite, (b) liberate ammonia from ammonium chloride, (c) not affect litmus? [S27, S26]

3.18 Mohan has solutions P, Q, R with pH 13, 5, 2. Which will (a) react with magnesium to liberate hydrogen, (b) liberate ammonia from NH_4Cl, (c) contain molecules as well as ions? [T24]

3.19 For which pH change is the increase in acidity maximum: 1→3, 2→6, 3→1, 6→2? [S26]

3.20 Which two statements confirm the acidic nature of H_2SO_4? (1) $BaCl_2$ gives a white precipitate (2) SO_2 evolves with carbon (3) pH is below 7 (4) CuO gives a blue solution [S27]

3.21 Which solution would dissociate to give a positive ion other than H_3O^+ and a negative ion other than OH^-: KOH, dil. HCl, NaCl or CH_3COOH? [S26]

D. Preparation of salts

3.22 List the six methods of preparing normal salts. Give one equation for each: direct combination, displacement, precipitation (double decomposition), neutralisation of an insoluble base, neutralisation of an alkali (titration), action of dilute acid on carbonates and bicarbonates.

3.23 Salts can be prepared by dilute acid acting on (1) bases (2) carbonates (3) metals (4) sulphites. Which method cannot be used for copper salts? [T26]

3.24 Using chemicals from this list, write balanced equations for: (a) lead(II) carbonate, (b) copper(II) chloride, (c) iron(II) chloride. List: lead nitrate, copper, hydrochloric acid, lead oxide, iron, sodium carbonate, copper carbonate, chlorine. [T26]

3.25 Which pair best produces lead(II) sulphate? (a) H_2SO_4 + Pb (b) H_2SO_4 + $Pb(OH)_2$ (c) Na_2SO_4 + $Pb(NO_3)_2$ (d) K_2SO_4 + PbO [T25]

3.26 Choose the method: P simple displacement, Q neutralisation by titration, R precipitation, S direct combination. Method for: (a) $PbCl_2$, (b) $FeCl_3$, (c) Na_2SO_4. [T25]

3.27 Match the salt with the best method: (a) $ZnCl_2$ from Zn, (b) KNO_3 from KOH, (c) $CaCO_3$ from $CaCl_2$. Methods: precipitation, direct combination, displacement, neutralisation. [T24]

3.28 The salt that can be prepared by direct combination: $FeCl_3$ or $FeCl_2$? [T24] A salt prepared by neutralisation involving titration: $CaCl_2$ or NH_4Cl? [S27]

3.29 Write equations for preparing: (a) copper sulphate from copper carbonate, (b) sodium zincate from zinc. [S26]

3.30 Convert NaOH to Na_2SO_4 using H_2SO_4 and write the balanced equation. [T23]

3.31 N is a metal of atomic number 11 that forms a basic oxide. Give (a) the formula of its oxide, (b) the equation when the oxide is treated with water, (c) the salt formed when sulphuric acid is added to the product in (b). [S27]

Answer Key

Objective and numerical items only. Write theory items from your own notes, then check them against your textbook.

Chapter 1

1.10 (c) 12

1.11 (c) A true, R false (the most reactive metal is at the bottom of Group 1)

1.12 Cl^- and K^+ (both 2, 8, 8)

1.13 Z

1.14 (d)

1.15 (d) Br

1.16 Li

1.17 (d) argon

1.18 Na

1.19 3 and 7

1.20 17

1.21 6

1.22 (c) X iodine, Y bromine

1.23 Yes, both true

1.24 (a) P (b) R (c) R > Q > P

1.25 (a) fluorine (b) sodium

1.26 (a) F (b) Ca (c) K

1.27 (a) Cl (b) Ne

1.28 (a) period 3 (b) XY_2

1.29 (a) Be > C > F (b) K < Na < Li

1.30 (a) hydrogen (b) sulphur (c) chlorine

1.31 (a) K > Ca > Mg > Al (b) Be < C < N < O (c) F > P > Si > Be

1.32 (a) Group 17, period 3 (b) less

1.33 (a) 2 (b) Group 2, alkaline earth metals (c) $Z(OH)_2$

1.34 (a) X has more (b) X is more electronegative (c) right

1.35 (a) M (b) N (c) L

1.36 largest

Chapter 2

2.13 R

2.14 sodium chloride

2.15 HCl; CCl_4

2.16 bad

2.17 CCl_4

2.18 (c) two lone pairs

2.19 H_3O^+

2.20 molten sodium chloride

2.21 Water: covalent compound; alkali metal: lithium; halogen: iodine; calcium oxide: ionic compound; weak acid: acetic acid

2.22 P_2Q (Na_2O); 2(23) + 16 = 62 g

2.23 (a) False, the cation attains the configuration of the noble gas that comes before it (b) a molten or dissolved ionic compound conducts through ions; in the solid state it does not

Chapter 3

3.8 (c) both true, R explains A

3.11 sodium sulphide

3.12 potassium sulphite

3.13 SO_2

3.14 iron and zinc (both give hydrogen); copper gives none, magnesium does

3.15 rotten-egg smelling gas (H_2S)

3.16 Q

3.17 (a) X (b) Z (c) Y

3.18 (a) Q and R (b) P (c) Q

3.19 6 → 2

3.20 statements 3 and 4

3.21 NaCl

3.23 action of dilute acid on metals (copper does not react with dilute acids)

3.24 (a) $Pb(NO_3)_2 + Na_2CO_3 \rightarrow PbCO_3 + 2NaNO_3$ (b) $CuCO_3 + 2HCl \rightarrow CuCl_2 + H_2O + CO_2$ (c) $Fe + 2HCl \rightarrow FeCl_2 + H_2$

3.25 (c)

3.26 (a) R (b) S (c) Q

3.27 (a) displacement (b) neutralisation (c) precipitation

3.28 $FeCl_3$; NH_4Cl

3.29 (a) $CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2$ (b) $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$

3.30 $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

3.31 (a) Na_2O (b) $Na_2O + H_2O \rightarrow 2NaOH$ (c) Na_2SO_4

Next banks: Chapter 4 (Analytical Chemistry) and Chapter 5 (Mole Concept) follow once you have cleared and logged gaps from this set.